

Part 1: Basic Discrete Probability

Example: A European call option with strike price K and maturity T gives the holder the right to purchase a share of said stock for K dollars at time T .

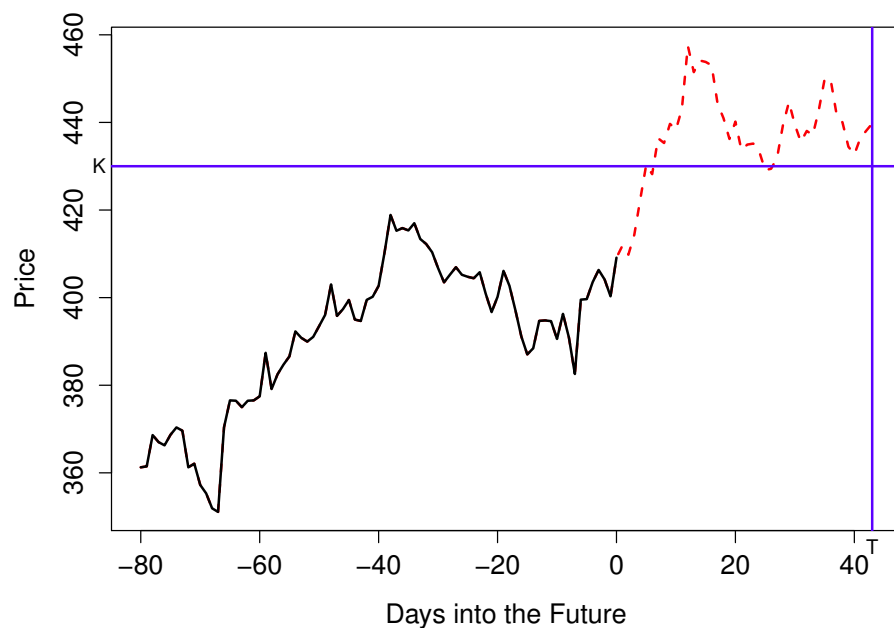


Figure 1: An example where $K = 430$ and $T = 43$. Time is adjusted so that the present is time zero. (This is GOOG from January through June, 2013.)

What is a reasonable premium to pay for this option?

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Probability Models are used in the place of the real processes which generate the data.

A key feature distinguishing these from **deterministic models** is that they allow for studying the variability in the system, and from this we can calculate expected values and the probabilities of events of interest:

$$P(A) = \text{the probability of event } A$$

A recurring challenge is to make probability models sufficiently simple to mathematically tractable while being sufficiently realistic to be useful.

Example: “Jumps” can be incorporated into a model for a stock price to make it more realistic than a model with continuous paths.

See, for example, Kou (2002).

A Jump-Diffusion Model for Option Pricing

S. G. Kou

Department of Industrial Engineering and Operations Research, 312 Mudd Building,
Columbia University, New York, New York 10027
kou@ieor.columbia.edu

KOU

A Jump-Diffusion Model for Option Pricing

Brownian motion and normal distribution have been widely used in the Black–Scholes option-pricing framework to model the return of assets. However, two puzzles emerge from many empirical investigations: the leptokurtic feature that the return distribution of assets may have a higher peak and two (asymmetric) heavier tails than those of the normal distribution, and an empirical phenomenon called “volatility smile” in option markets. To incorporate both of them and to strike a balance between reality and tractability, this paper proposes, for the purpose of option pricing, a double exponential jump-diffusion model. In particular, the model is simple enough to produce analytical solutions for a variety of option-pricing problems, including call and put options, interest rate derivatives, and path-dependent options. Equilibrium analysis and a psychological interpretation of the model are also presented.

(Contingent Claims; High Peak; Heavy Tails; Interest Rate Models; Rational Expectations; Overreaction and Underreaction)

1. Introduction

Despite the success of the Black–Scholes model based on Brownian motion and normal distribution, two empirical phenomena have received much attention recently: (1) the asymmetric leptokurtic features—in other words, the return distribution is skewed to the left, and has a higher peak and two heavier tails than those of the normal distribution, and (2) the volatility smile. More precisely, if the Black–Scholes model is correct, then the implied volatility should be constant. In reality, it is widely recognized that the implied volatility curve resembles a “smile,” meaning it is a convex curve of the strike price.

Many studies have been conducted to modify the Black–Scholes model to explain the two empirical phenomena. To incorporate the asymmetric leptokurtic features in asset pricing, a variety of models have been proposed:¹ (a) chaos theory,

fractal Brownian motion, and stable processes; see, for example, Mandelbrot (1963), Rogers (1997), Samorodnitsky and Taqqu (1994); (b) generalized hyperbolic models, including log t model and log hyperbolic model; see, for example, Barndorff-Nielsen and Shephard (2001), Blattberg and Gonedes (1974); (c) time-changed Brownian motions; see, for example, Clark (1973), Madan and Seneta (1990), Madan et al. (1998), and Heyde (2000). An immediate problem with these models is that it may be difficult to obtain analytical solutions for option prices. More precisely, they might give some analytical formulae for standard European call and put options, but any analytical solutions for interest rate derivatives and path-dependent options, such as perpetual American options, barrier, and lookback options, are unlikely.

In a parallel development, different models are also proposed to incorporate the “volatility smile” in option pricing. Popular ones include: (a) stochastic volatility and ARCH models; see, for example, Hull and White (1987), Engle (1995), Fouque et al. (2000); (b) constant elasticity model (CEV) model; see,

for example, Cox and Ross (1976), and Davydov and Linetsky (2001); (c) normal jump models proposed by Merton (1976); (d) affine stochastic-volatility and affine jump-diffusion models; see, for example, Heston (1993), and Duffie et al. (2000); (e) models based on Lévy processes; see, for example, Geman et al. (2001) and references therein; (f) a numerical procedure called “implied binomial trees”; see, for example, Derman and Kani (1994) and Dupire (1994). Aside from the problem that it might not be easy to find analytical solutions for option pricing, especially for path-dependent options (such as perpetual American options, barrier, and lookback options), some of these models may not produce the asymmetric leptokurtic feature (see §2.3).

The current paper proposes a new model with the following properties: (a) It offers an explanation for two empirical phenomena—the asymmetric leptokurtic feature, and the volatility smile (see §§3 and 5.3). (b) It leads to analytical solutions to many option-pricing problems, including European call and put options (see §5); interest rate derivatives, such as swaptions, caps, floors, and bond options (see §5.3 and Glasserman and Kou 1999); path-dependent options, such as perpetual American options, barrier, and lookback options (see §2.3 and Kou and Wang 2000, 2001). (c) It can be embedded into a rational expectations equilibrium framework (see §4). (d) It has a psychological interpretation (see §2.2).

The model is very simple. The logarithm of the asset price is assumed to follow a Brownian motion plus a compound Poisson process with jump sizes double exponentially distributed. Because of its simplicity, the parameters in the model can be easily interpreted, and the analytical solutions for option pricing can be obtained. The explicit calculation is made possible partly because of the memoryless property of the double exponential distribution.

The paper is organized as follows. In §2, the model is proposed, is evaluated by four criteria, and is compared with other alternative models. Section 3 studies the leptokurtic feature. A rational expectations equilibrium justification of the model is given in §4. Some preliminary results, including the Hh functions, are given in §5.1. Formulae for option-pricing problems, including options on futures, are provided in §5.2.

The “volatility smiles” phenomenon is illustrated in §5.3. The final section discusses some limitations of the model.

2. The Model

2.1. The Model Formulation

The following dynamic is proposed to model the asset price, $S(t)$, under the physical probability measure P :

$$\frac{dS(t)}{S(t-)} = \mu dt + \sigma dW(t) + d\left(\sum_{i=1}^{N(t)} (V_i - 1)\right), \quad (1)$$

where $W(t)$ is a standard Brownian motion, $N(t)$ is a Poisson process with rate λ , and $\{V_i\}$ is a sequence of independent identically distributed (i.i.d.) nonnegative random variables such that $Y = \log(V)$ has an asymmetric double exponential distribution^{2,3} with the density

$$f_Y(y) = p \cdot \eta_1 e^{-\eta_1 y} 1_{[y \geq 0]} + q \cdot \eta_2 e^{\eta_2 y} 1_{[y < 0]},$$

$$\eta_1 > 1, \eta_2 > 0,$$

where $p, q \geq 0, p + q = 1$, represent the probabilities of upward and downward jumps. In other words,

$$\log(V) = Y \stackrel{d}{=} \begin{cases} \xi^+, & \text{with probability } p \\ -\xi^-, & \text{with probability } q \end{cases}, \quad (2)$$

where ξ^+ and ξ^- are exponential random variables with means $1/\eta_1$ and $1/\eta_2$, respectively, and the notation $\stackrel{d}{=}$ means equal in distribution. In the model, all sources of randomness, $N(t)$, $W(t)$, and Y s, are assumed to be independent, although this can be relaxed, as will be suggested in §2.2. For notational simplicity and in order to get analytical solutions for various option-pricing problems, the drift μ and the volatility σ are assumed to be constants, and the Brownian motion and jumps are assumed to be

¹ Although most of the studies focus on the leptokurtic features under the physical measure, it is worth mentioning that the leptokurtic features under the risk-neutral measure(s) lead to the “volatility smiles” in option prices.

² If $\eta_1 = \eta_2$ and $p = 1/2$, then the double exponential distribution is also called “the first law of Laplace” (proposed by Laplace in 1774), while the “second law of Laplace” is the normal density.

³ Ramezani and Zeng (1999) independently propose the same jump-diffusion model from an econometric viewpoint as a way of improving the empirical fit of Merton’s normal jump-diffusion model to stock price data.

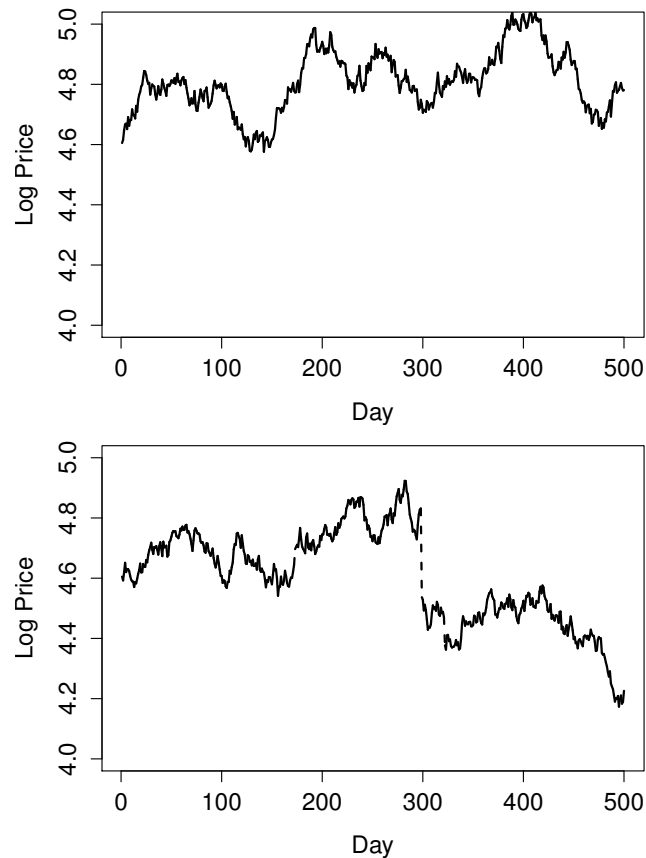


Figure 2: Examples of Brownian motion paths from a processes with and without jumps.

Comments on Kou:

1. Note how the author balances realism and mathematical practicality with this choice of model.
2. **Model validation** is a crucial step, consisting of comparing your model with real data.
3. Note the importance of understanding a wide range of probability distributions.

Exercise: One (of multiple) components to this model is the distribution that specifies the jump sizes. One puts forth a model in the form of

The size of a jump (denote it X) is modelled as a random variable with density $f(\cdot)$, i.e.,

$$P(a \leq X \leq b) = \int_a^b f(u) du, \quad a \leq b.$$

What assumption does Kou make regarding $f(\cdot)$? Does this assumption, by itself, enable the calculation of all probabilities regarding jumps?

Formally, a probability model consists of three parts:

1. The **sample space** (Ω), the collection of all possible **outcomes** for the probability experiment.
2. A collection of subsets of Ω , called the **events** and denoted $\mathcal{F}$, for which the probability is defined.
3. The **probability measure** (P) assigns probabilities to elements of $\mathcal{F}$.

Often write $(\Omega, \mathcal{F}, P)$ as the **the probability space**.

Exercise: Does $\mathcal{F}$ contain every subset of Ω ?

Assigning Probabilities to Events

In some cases, $P(A)$ is “known” or “given” or “obvious.” (But, be careful with your intuition.)

In other cases, $P(A)$ is found using “rules.”

However it is done, the probability measure must satisfy the following:

1. $P(A) \geq 0$ for all $A \in \mathcal{F}$

2. $P(\Omega) = 1$

3. If $A_1, A_2, \dots$, are disjoint events ($A_i \cap A_j = \emptyset$ for $i \neq j$), then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

Set-Theoretic Probability Rules

1. **Equally-Likely Outcomes:** If Ω is a finite collection of outcomes and these outcomes are **equally-likely**, then $\mathcal{F} = 2^\Omega$ and for any $A \in \mathcal{F}$,

$$P(A) = \frac{\text{number of elements in } A}{\text{number of elements in } \Omega}.$$

2. **The Addition Rule:** For any events A, B ,

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

3. **The Rule for Complements:** For any event A ,

$$P(A^c) = 1 - P(A).$$

Exercise: Suppose I toss a (fair, six-sided) die twice. What is the probability that the sum of the two tosses is seven?

Exercise (From *Heard on the Street*, 12th Edition, by Crack): You are dealt exactly two playing cards from a well-shuffled standard 52-card deck. The standard deck contains exactly four Kings. What is the probability that both of your cards are Kings?

1 **Exercise (From HOTS):** Four cards are shuffled and placed face down in
2 front of you. Their faces (hidden) display the four elements: water, earth,
3 wind and fire. You are to turn the cards over, one at a time, until you either
4 win or lose. You win if you turn over water and earth. You lose if you turn
5 over fire. What is the probability that you win?

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1 **Exercise (From HOTS):** Take a stick and break it randomly into three pieces
2 (i.e., two randomly placed breaks on the stick). What is the probability you
3 can form a triangle from the pieces?

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Conditional Probability Rules

1. **Rule for Conditional Probability:** For any events A, B with $P(B) > 0$,

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

2. **Multiplication Rule:** For any events A, B ,

$$P(A \text{ and } B) = P(A \cap B) = P(A|B)P(B).$$

3. **Law for Total Probability:** Suppose events $B_1, B_2, \dots, B_n$ form a **partition** of Ω . Then for any event A ,

$$P(A) = \sum_{i=1}^n P(A|B_i)P(B_i).$$

4. **Bayes' Theorem:** Suppose events $B_1, B_2, \dots, B_n$ form a partition of Ω . Then for any event A ,

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum_{j=1}^n P(A|B_j)P(B_j)}.$$

- Exercise (Partially from HOTS):** You are dealt exactly two playing cards from a well-shuffled standard 52-card deck. The standard deck contains exactly four Kings. (a) What is the probability that both of your cards are Kings? (b) What is the probability that the second card you are dealt is a King?

Exercise (From HOTS): You have a large jar containing 999 fair pennies and one two-headed penny. Suppose you pick one coin out of the jar and flip it ten times and get all heads. What is the probability that the coin you chose is the two-headed one?

Exercise (Similar to question in HOTS): Polygraph tests (lie detector tests) are often routinely administered to employees or prospective employees in sensitive positions. According to studies of polygraph reliability (Gastwirth, J., 1987. The statistical precision of medical screening procedures, *Statistical Science*, 3, 213-222) if a person is lying, the probability that this is detected by the polygraph is 0.88, whereas if the person is telling the truth, the polygraph is indicating that he/she is telling the truth with probability 0.86. Now suppose that on a particular question the vast majority of subjects have no reason to lie, so that 99% tell the truth.

Suppose a subject produces a positive reading on the polygraph (hence indicating that he/she is lying). What is the probability that the polygraph is incorrect and that he/she is telling the truth?

Exercise (Very similar to a question in HOTS): I flip two coins hidden from your view.

Version One: You ask, "Is at least one of the coins heads?" I respond, "Yes."

What is the probability they are both heads (from your perspective)?

Version Two: I say "At least one is heads." What is the probability they are both heads (from your perspective)?

Version Three: You happen to see one of the coins, and notice that it is heads.

What is the probability they are both heads (from your perspective)?

22 horizontal lines for writing answers.

21 horizontal lines for writing answers.

Exercise (From HOTS): This is one version of the famous “Let’s Make a Deal” or “Monty Hall” game show problem. There are three doors. You know that there is a prize behind one of them, and nothing behind the other two. The game show host tells you that you shall receive whatever is behind the door of your choice. However, before you choose, he promises that rather than immediately opening the door of your choice to reveal its contents, he will open one of the other two doors to reveal that it is empty. He will then give you the option to switch your choice. You may assume that the host is totally impartial – not malicious in any way. You choose Door 3. He opens Door 2 and reveals that it is empty. You now know that the prize lies behind either Door 3 or Door 1. Should you switch your choice to Door 1?

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Parade Picks

Edited by Vi-An Nguyen / LIKE US AT [FACEBOOK.COM/PARADEMAG](https://www.facebook.com/parademag)

IDAHO
Forest and conservation technicians

NEW JERSEY
Marriage and family therapists

HAWAII
Dancers

TENNESSEE
Nuclear Technicians

THE MOST POPULAR JOB IN EVERY STATE

These four states are hotspots for the occupations above, according to the U.S. Bureau of Labor Statistics*. You'll never guess which job is unusually popular in Alaska! Check out a chart of all 50 states' top jobs at parade.com/labor.

*The jobs have the highest relative concentration in their respective states.



For National Preparedness Month, America's favorite heroic dog is teaming up with Save the Children to spread the word about the importance of emergency kits for kids. From medical info to recent photos, learn what should go in every kit—and see how Lassie saves the day when she drops by the *Parade* offices—at parade.com/lassie.

CLASH OF THE TITANS

As the NFL season kicks off this week, we asked the sportscasters of NBC's *Sunday Night Football* and *Football Night in America* which big games they're most excited to watch. Share yours at parade.com/NFL.



AL MICHAELS and **CRIS COLLINSWORTH**: San Francisco at Denver (Oct. 19) "This is a possible Super Bowl preview," Michaels says. "These are two of last year's best teams, with star QBs—one at the end of a great career [Peyton Manning] and one at the start [Colin Kaepernick]," Collinsworth adds.



BOB COSTAS: Chicago at Green Bay (Nov. 9) "All of the matchups look good. But the Bears and the Packers are the most-played rivalry in the league," Costas says.



MICHELE TAFOYA and **HINES WARD**: Seattle at San Francisco (Thanksgiving) "Hands down, it's the NFC Championship Game rematch—a fierce rivalry with exciting quarterbacks and loads of fascinating personalities," Tafoya says. "These teams have genuine dislike but the utmost respect for each other," Ward says.



Ask Marilyn

By Marilyn vos Savant

Say that 1 million scratch-off cards are sold. Under the silver-coated area, one card reads, "You win a date with the man or woman of your dreams." All the other cards read, "You lose." After buying a card for myself, I observe 999,998 other contestants scratch off their cards to read, "You lose."

Words We Need

izzat (interjection) reflexive comment made upon spotting a celebrity

A 999,999th contestant, his unscratched card still in his back pocket, is playing Frisbee with his dog. Should I offer him a lot of money to swap scratch-off cards with me? —B. H., San Jose, California
No, because your chance of winning (now 50-50) wouldn't improve. Instead, offer him a lot of money to sell his card to you.

Numbrix

Complete 1 to 81 so the numbers follow a horizontal or vertical path—no diagonals.

3	5	43	49	51
9				77
11				75
21				69
29	31	33	63	67

Part 1: Basic Discrete Probability

Exercise: It is difficult to conduct sample surveys on sensitive issues because many people will not answer questions if the answers might be embarrassing or incriminating. *Randomized Response* is an effective way to guarantee anonymity while collecting information on topics such as student cheating or tax evasion. (Warner, *Journal of the American Statistical Association*, 1965).

For example, to ask a sample of students whether they have ever used illegal drugs, have each student toss a fair coin *in private*. If the coin lands **Heads**, they are to tell the **truth**. If the coin lands **Tails** they are to answer **"Yes"** (regardless of the truth). Only the student knows whether the "embarrassing" answer ("Yes") reflects the truth or just the coin toss.

Explain how the randomized response method can be used to estimate the probability that a student has used illegal drugs.

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Independent Events

A pair of events A, B are defined to be **independent** if $P(A \cap B) = P(A)P(B)$.

A sequence of events $A_1, A_2, \dots, A_n$ are independent if for **every** collection of indices

$$\{i_1, i_2, \dots, i_k\} \subseteq \{1, 2, \dots, n\}$$

for $k \leq n$ it holds that

$$P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}) = \prod_{j=1}^k P(A_{i_j})$$

Comment: Note that for a sequence of events $A_1, A_2, \dots, A_n$ it is **not** sufficient that

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = \prod_{i=1}^n P(A_i)$$

in order that $A_1, A_2, \dots, A_n$ be independent.

But, of course, if $A_1, A_2, \dots, A_n$ are **assumed** to be independent, then it holds that

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = \prod_{i=1}^n P(A_i)$$

This is the most commonly used application of independent events.



15sigma22pa

KNOW AND SHOW 5

[CLICK HERE for Know And Show GLOSSARY of TERMS](#)

Category 6: Chances Are

Three out of 45 games that Arlene bowls, she scores a minimum of 280 points. How would you describe her chances that on her next 18 games she will bowl at least one game with a score of 280 points or higher?

Points Earned: 0

click answer

Good	☐ A
Very Poor	☐ B
Poor	☐ C
Very Good	☐ D

Part 1: Basic Discrete Probability

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Exercise (From HOTS): If you like gambling and you like betting on the outcome of sports matches, then you may like the “parlay card.” A parlay card lets you bet on the outcomes of more than one match. In order to win a parlay bet, you must be correct on each of the matches you bet upon. Parlay cards offer big payoffs if you are right on every match.

Suppose that your bookie will give you 10-to-1 odds for a parlay bet that covers four sports matches. Should you take the bet?

[illegible]

Conditional Independence

Suppose that events A_1 and A_2 are not necessarily independent, but there is another event B such that

$$P(A_1 \cap A_2|B) = P(A_1|B)P(A_2|B)$$

Then we say that A_1 and A_2 are **independent conditional on B** .

Example: Imagine that there are three coins: One is a regular two-sided coin, another has “Heads” on both sides, and another has “Tails” on both sides. One of three coins is chosen at random, and then flipped twice. Let A_i be the event that flip i comes up “Heads.”

Then, A_1 and A_2 are dependent, but if I define B_j to be the event that the chosen coin has j sides with “Heads,” (with $j = 0, 1, 2$), then A_1 and A_2 are independent conditional on any one of the B_j .

Exercise: A bank classifies customers as either a good or bad credit risk. On the basis of extensive historical data, the bank has observed that 1% of good credit risks and 10% of bad credit risks overdraw their account in any given month. A new customer opens a checking account at the bank. On the basis of a check with the credit bureau, the bank believes that there is a 70% chance that the customer will turn out to be a good credit risk.

Suppose that this customer's account is overdrawn in the first month. How does this alter the bank's opinion of this customer's creditworthiness? What if the customer overdraws the account again in the second month?
